EEG Signal Analysis for Epileptic Seizure Detection with Deep Learning Approaches

Ivana Horvat
Machine Learning for Medicine
Faculty of Technical Sciences
Novi Sad, Serbia

Abstract—**Epilepsy is a chronic neurological disorder affecting approximately 50 million people worldwide, with misdiagnosis rates estimated at 20 to 30 percent. Accurate automated seizure detection from electroencephalography (EEG) signals is a clinically important open problem. This work evaluates two deep learning architectures - Baseline Convolutional Neural Network (CNN) and ResNet18 - for binary epileptic seizure detection on the UCI Epileptic Seizure Recognition Dataset. One-dimensional EEG signal was transformed into a 2D time-frequency scalogram using Continuous Wavelet Transform with a Morlet wavelet. Both models were evaluated under two optimisation strategies: maximising F1-score and recall. ResNet18 optimised for F1-score achieved the best overall performance, with an F1-score of 94.6%, precision of 100%, and recall of 89.8%. The Baseline CNN optimised for F1-score achieved an F1-score of 93.2%, precision of 99.4%, and recall of 87.7%. Recall optimisation did not meaningfully improve sensitivity for the Baseline CNN, while for ResNet18 it reduced missed seizures at the cost of additional false positives. These findings suggest that ResNet18 with F1 optimisation provides the most reliable performance for this task, while the trade-off between false negatives and false positives remains a key consideration for clinical conditions.**

Keywords—epileptic seizure detection; EEG signal classification; Convolutional Neural Networks; Resnet 18

I. Introduction

Epilepsy is a chronic neurological disorder characterized by recurrent seizures caused by abnormal electrical activity in the brain. It affects approximately 50 million people worldwide, making it one of the most common neurological conditions globally [1]**Error! Reference source not found.**. Accurate and timely diagnosis is essential for effective treatment and prevention of complications associated with seizure episodes.

Electroencephalography (EEG) is one of the primary tools used in the diagnosis and monitoring of epilepsy [2]. EEG signals represent the electrical activity of the brain measured over time and provide valuable insight into neural dynamics. However, these signals are very complex, non-linear, and highly variable across patients and conditions. EEG recordings often contain noise, artifacts, and overlapping patterns, making their interpretation challenging even for experienced clinicians**Error! Reference source not found.** [3].

In clinical practice, EEG signals are typically analyzed manually by neurologists. This process is time-consuming and subject to human error, particularly given the large volume of data and the subtle differences between seizure and non-seizure patterns. Studies indicate that epilepsy may be misdiagnosed in up to 20–30% of cases, which can lead to inappropriate treatment, increased health risks, and reduced quality of life for patients [4]. These limitations highlight the need for automated and reliable seizure detection systems.

The main challenge in EEG-based seizure detection lies in the nature of the signal itself. Seizure-related patterns are not always localized or easily distinguishable in the time domain, and important information is often distributed across both temporal and frequency components beyond what traditional analysis methods can capture [3].

In recent years, machine learning and deep learning approaches have been increasingly applied to EEG signal analysis [5][6]. These methods enable automatic feature extraction and can model complex, non-linear relationships within the data. Classical machine learning models, such as Support Vector Machines (SVM) and Random Forests (RF), have shown promising results when combined with appropriate preprocessing techniques [7]. More advanced deep learning architectures further improve performance by capturing spatial and temporal dependencies in EEG signals [5].

This work evaluates two CNN architectures - a three-block Baseline CNN and ResNet18 - for binary seizure detection on EEG scalograms derived from the UCI Epileptic Seizure Recognition Dataset. Each architecture was trained and evaluated under two optimisation strategies, targeting F1-score and recall separately, to investigate the clinical trade-off between overall performance and seizure sensitivity.

The main contribution of this project is the evaluation of ResNet18 architecture for epilepsy detection using a well-known publicly available dataset. The study investigates the impact of preprocessing techniques such as standardization and scalogram generation and how this impacts the model performance.

The remainder of the technical report is organised as follows: Section II describes related work, Section III presents the methodology, Section IV discusses experimental results, and Section V concludes the report.

II. Related work

Several studies have investigated automated epileptic seizure detection from EEG signals, employing a range of signal processing, machine learning, and deep learning approaches. This section provides an overview of the most relevant work, grouped by methodology.

Early approaches to automated seizure detection relied on conventional signal processing combined with manual feature extraction. Techniques such as Fourier Transform and Wavelet Transform were commonly used to move EEG signals from the time domain into the time-frequency domain, after which extracted features were passed to classifiers such as Support Vector Machines (SVM), K-Nearest Neighbours (KNN), or Random Forests [3]. These methods achieved reasonable classification accuracy. More recently, Atlam et al. [7] proposed a hybrid pipeline combining Discrete Wavelet Transform (DWT) and Principal Component Analysis (PCA) for feature extraction, alongside SMOTE for class balancing, achieving 97.3% accuracy and an F1-score of 93.08% using SVM on the UCI Epileptic Seizure Recognition Dataset.

In recent years, deep learning approaches have increasingly replaced manual feature extraction by learning representations directly from data. Shoeibi et al. [5] provide a comprehensive review of deep learning techniques applied to seizure detection, covering Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and hybrid architectures and discusses advantages and limitations of deep learning based methods for automated seizure detection.

A particularly relevant line of work involves converting EEG signals into two-dimensional time-frequency images prior to classification. Bizopoulos et al. [8][8] demonstrated that representing EEG as scalogram images and feeding them into deep neural networks produces strong classification performance, coining the term Signal2Image modules to describe this approach. Türk and Özerdem [3] applied Continuous Wavelet Transform (CWT) with a Morlet wavelet to generate scalograms from EEG recordings and trained a CNN classifier, achieving up to 99.5% accuracy on binary classification tasks. Dev et al. [9] further validated the scalogram-based approach by combining Morlet CWT scalograms with hybrid deep learning models for automated seizure detection, demonstrating strong performance and supporting the use of CWT scalograms as an effective EEG representation for deep learning pipelines.

A recurring methodological concern in literature is data leakage arising from segment-level train-test splitting. When EEG recordings from the same subject appear in both training and test sets, models may learn subject-specific characteristics rather than generalisable seizure patterns, leading to inflated performance estimates [10]. This concern motivates the subject-level splitting strategy adopted in the present work.

The present study builds on the scalogram-based CNN approach established in the literature, extending it in two directions. First, two architectures are compared - a simple three-block Baseline CNN and the deeper ResNet18 - to assess whether added architectural complexity yields meaningful performance gains on this dataset. Second, each model is evaluated under two optimisation strategies, targeting F1-score and recall separately, to explicitly examine the clinical trade-off between overall classification performance and seizure sensitivity.

III. Methodology

This section describes the full pipeline used to train and evaluate models compared in this work. It begins with an exploratory analysis of the dataset, followed by the preprocessing steps applied before training. The model architectures are then described, along with the hyperparameter optimization procedure. The section closes with the experimental setup description.

A. *Exploratory Data Analysis*

This study utilizes the publicly available UCI Epileptic Seizure Recognition Dataset, widely used in EEG-based seizure detection research, available online on the platform Kaggle [11].

An exploratory data analysis (EDA) was conducted to understand the structure and statistical properties of the dataset. The analysis included inspection of feature distributions, class balance, and visualization of EEG signals across classes. Statistical measures such as mean, variance, and standard deviation were computed for each class.

The dataset consists of 11,500 EEG signal segments collected from 500 subjects. Each subject contributes 23 segments, where each segment represents approximately one second of EEG recording. Each instance is described by 178 numerical features corresponding to discrete samples of the signal, along with a class label.

Seizure segments differ from non-seizure segments in the variability of the signal. The seizure class exhibited higher variability, which is consistent with the nature of epileptic EEG activity. Outlier analysis indicated the presence of extreme values, however, these were retained as they represent the nature of the EEG signal rather than noise (Fig. 1).

The dataset was examined for missing values and none were detected.

The dataset contains five classes representing different physiological conditions of brain activity:

- Class 1: EEG signals recorded during epileptic seizure (ictal)
- Class 2: EEG signals recorded before seizure onset (preictal)
- Class 3: EEG signals recorded between seizures (interictal)
- Class 4: EEG signals from healthy subjects with eyes open
- Class 5: EEG signals from healthy subjects with eyes closed

Classes 1–3 correspond to subjects diagnosed with epilepsy, while classes 4–5 represent healthy individuals. Only Class 1 corresponds to active seizure activity, characterized by increased signal amplitude and frequency, while all other classes represent non-seizure brain states.

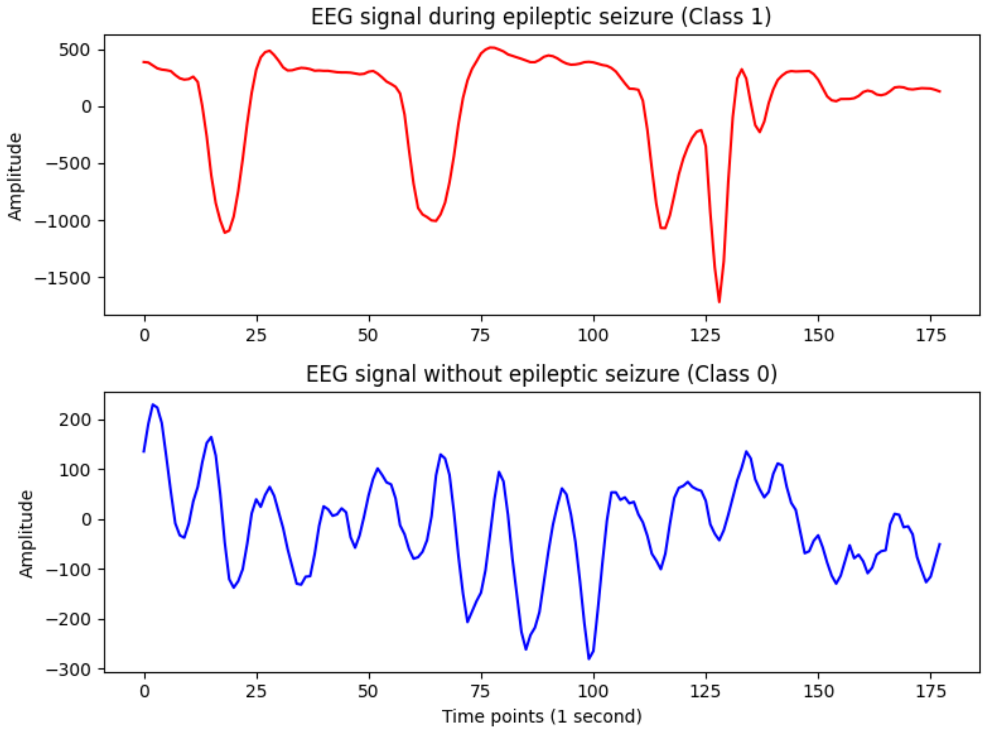

Fig. 1. EEG signal: during epileptic seizure (above) and without seizure (below)

B. Data Preprocessing

1) Binarization

For this study, the problem is reformulated as a binary classification task:

- Seizure (Class 1)
- Non-seizure (Classes 2–5)

This transformation results in an imbalanced dataset, with 20% seizure samples and 80% non-seizure samples. The target variable has value 1 for seizure (original class 1) and 0 for non-seizure (original classes 2–5).

2) Standardization

Feature standardization was performed using the StandardScaler method. The scaler was fitted exclusively on the training data to compute mean and standard deviation, and the learned parameters were applied to the validation and test sets, ensuring there is no data leakage between the data subsets.

3) Scalogram generation

Prior to model training, each 1D EEG signal was transformed into a 2D time-frequency representation using the Continuous Wavelet Transform (CWT) with a Morlet wavelet (Fig. 2). The Morlet wavelet was selected as it provides well-balanced resolution in both the time and frequency domains, which is well suited for oscillatory signals such as EEG [3][9][9]. For each signal, CWT was applied across 178 scales, yielding a scalogram of shape (178, 178) - a 2D matrix in which each row corresponds to a distinct frequency level and each column to a time point (for 1 second). Each cell in this matrix encodes the magnitude of wavelet coefficients, reflecting how strongly a given frequency was present at a given moment in time. The resulting scalograms were stored as single-channel images of shape (1, 178, 178) in NumPy format, enabling reuse across training runs.

4) Train, Validation, Test data split

The first column of the dataset encodes the recording identifier, with each unique recording containing exactly 23 one-second segments. We extract this identifier and use it as the grouping key for the train/validation/test split. Random sampling across rows of data would place segments of the same patient into different split sets. This would allow the model to memorize subject-specific patterns and impair its ability to generalize well. By partitioning the data by recording identifiers, we avoid data leakage [10].

The split is performed with *GroupShuffleSplit* from scikit-learn, using the extracted recording identifier as the grouping key. The resulting partition is:

- 70% training set
- 15% validation set
- 15% test set

Rows missing the recording identifier account for 1% of the dataset and were discarded before the split.

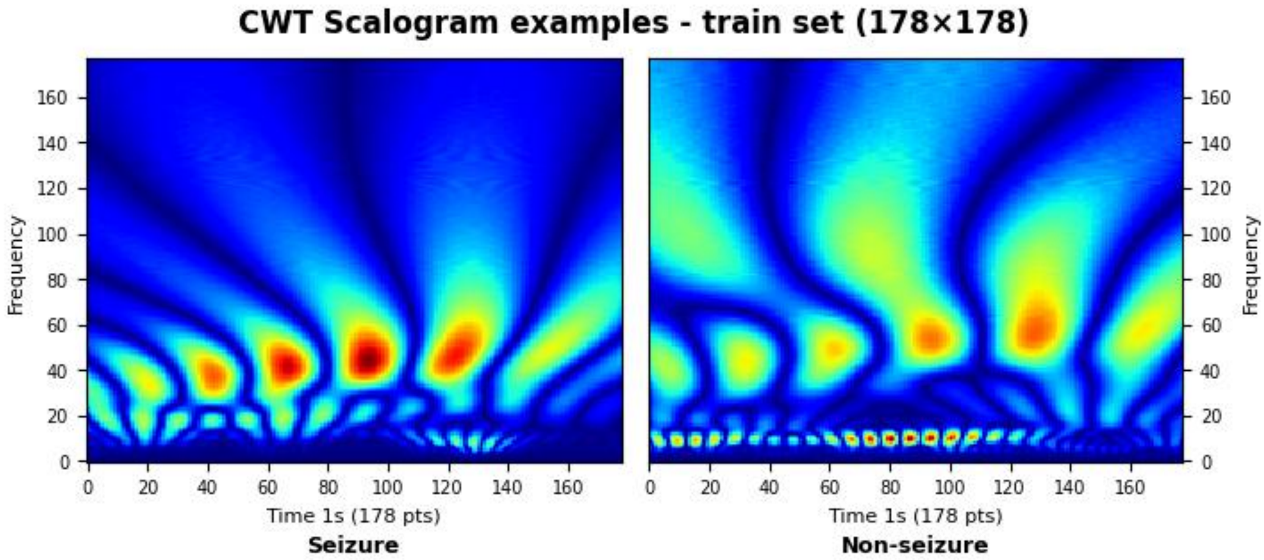

Fig. 2. Continous Wavelet Transformation applied to the EEG signal for two classes: sezier and non-seizure

C. Deep Learning Models

1) Baseline Convolutional Neural Network model

As the baseline, a three-block Convolutional Neural Network (CNN) model has been selected (Fig. 3). The input to the model is a scalogram of a shape (1, 178, 178). All convolutions use 3x3 kernels with padding=1 to preserve spatial dimension. The number of filters increases from 32 to 64 to 128 from the first to the third convolutional block. A Batch normalisation is applied to normalize the values. The ReLU activation function zeros out negative values. After that, Max Pooling is applied, halving the spatial size so each convolutional block looks at the larger patterns at lower resolution. The output of the final convolutional layer is a 3D feature map of shape (128, 22, 22). Flattening collapses it into a single vector of 61,952 numbers (128x22x22). After that, fully connected layer maps this vector to 256 neurons, ReLU activates them, while Dropout randomly disables neurons during training to prevent overfitting. The final linear layer outputs a single raw value. During evaluation, a sigmoid function converts this to a probability between 0 and 1. This architecture was selected as a baseline due to its sequential, non-residual structure, which makes it a suitable reference point for comparison against more complex architectures such as ResNet18, described in the following section.

2) ResNet 18

ResNet18 is a Convolutional Neural Network architecture consisting of 18 layers, 17 convolutional and one fully connected (FC) layer [12]. ResNet18 employs skipped connection method that addresses the degradation problem which emerges when neural networks become deeper. Skip connections preserve the original input by passing it directly to the output of the block, allowing the layers to learn only the residual - the additional information needed to improve the representation and make it more useful for the final classification.

For this project, the EEG scalograms were used as the input. Since ResNet18 expects 3-channel input for RGB images, it was modified to accept the 1-channel scalogram image of 178x178 pixels that was generated from the EEG signal recorded from patients. The model was trained from scratch, without pre-trained weights, as the dataset ResNet18 was originally trained on ImageNet, a large-scale dataset of everyday photographic images. ResNet18 is a well-established architecture with demonstrated performance on image classification tasks. For purpose of this paper, we tested how ResNet architecture performs on a domain-specific dataset such as EEG scalograms. Dropout was added before the fully connected layer to improve generalisation on the limited training set (~8000 training samples) [13] This allows direct comparison between a simple sequential architecture and a deeper residual network on the same task and dataset.

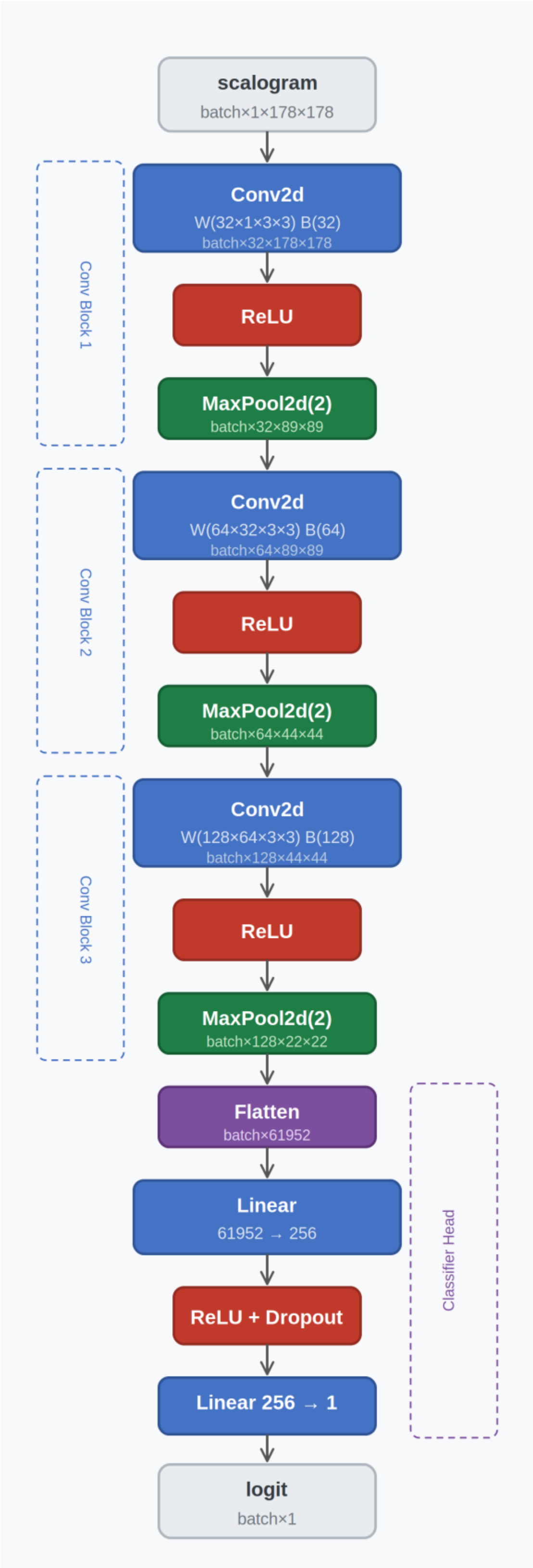

Fig. 3: Baseline CNN Architecture

D. Evaluation Metrics

Five metrics are used to evaluate all models: precision, recall, specificity, AUC/ROC and F1-score. Accuracy wasn't measured for this purpose as the dataset is highly imbalanced (only 20% of the dataset is class 1 – seizure). The confusion matrix is used to show prediction counts, providing a detailed overview of true positive (TP), false positive (FP), true negative (TN), and false negative (FN) predictions.

During model development, metrics were computed on the test set to monitor performance and models' generalization ability.

E. Hyperparameter Optimisation

Hyperparameter optimisation is the process of searching for the best combination of model parameters - those not learned during training - to maximise model performance. Hyperparameter optimisation was performed using Optuna [14], a Python-based automated optimization framework, with a random search sampler across 30 trials. The following hyperparameters were included in the search space: learning rate, batch size, dropout rate, and a positive class weight to counterweight the class imbalance.

Each trial used 5-fold cross-validation on the training set, with up to 25 epochs per fold and early stopping if validation loss did not improve for five consecutive epochs. Models were optimized for both F1-score and recall, and their performance was then compared. To reduce computational costs, a Median Pruner was implemented. At each step during a trial, the pruner compared the current trial's validation loss against the median of all completed trials. If the trial performed below the median, it was terminated early and a new trial began. The first five trials were exempt from this process to allow sufficient history of results to accumulate.

The best hyperparameter combination was used to retrain each model on the full training set running up to 50 epochs, using a validation set for early stopping with patience 10. The final evaluation of model performance was done on the test set. TABLE I. contains hyperparameter values used for Random Search optimization, across both models.

TABLE I. RANDOM SEARCH - MODEL HYPERPARAMETERS

Hyperparameter	**Baseline CNN & ResNet18**
Batch size	32, 64, 128
Pos_weight	75%, 100% and 125% of class ratio (4.41)
Learning rate	Log uniform between 1×10^{-4} and 1×10^{-2}
Dropout	0.3, 0.5, 0.7

F. Experimental Setup

Data preprocessing and analysis were implemented using the numpy, pandas, scikit-learn, and PyWavelets libraries. Scalogram generation was performed using the PyWavelets library via continuous wavelet transform. The neural network models were implemented in PyTorch with torchvision providing the ResNet18 architecture. Hyperparameter optimisation was conducted using Optuna. Results were visualised using matplotlib and seaborn.

The experiments were conducted on a personal computer running Windows 11 (64-bit), equipped with an NVIDIA GeForce RTX 3090 GPU with 24 GB VRAM. A fixed random seed (random_state = 42) was used to ensure reproducibility.

IV. RESULTS AND DISCUSSION

This section discusses results from experiments that are presented and compared across two model architectures. The discussion at the end of the section addresses the proposed solution's limitations and potential directions for further research. The goal of the experiments is to detect epileptic seizures from segments of EEG signals using supervised machine learning. The performance of a simple Baseline CNN is compared to the more complex ResNet18 architecture to examine how model complexity affects classification performance on the epileptic seizure detection task. Given the medical context, particular emphasis is placed on recall and F1-score, as minimising false negatives (missed seizures) is critical [2][4]. Both models were optimised separately for recall and F1-score, and the results are compared across both optimisation strategies.

A. Results

1) Random Search and Compute Time Results

This section provides a tabular overview of random search results for both models - Baseline CNN and ResNet18 - each optimised for two metrics: recall and F1-score. This gives a solid starting point for comparing models and the effect of different optimisation targets. Further details are available in Table II. Compute time for each model run is also reported, which will inform the overall assessment of model efficiency alongside performance.

2) Result comparison for Baseline CNN model for F1-score and Recall search optimisation

The two Baseline CNN variants - optimised for F1-score and recall respectively - are compared first. Both achieved identical recall of 87.7%, meaning the number of false negatives was equal in both cases (48 cases). However, the recall-optimised model produced seven times more false positives (14 vs 2). This indicates that optimising for recall made no meaningful difference to sensitivity but actively hurt precision. This suggests that Baseline CNN may have reached its performance ceiling given the model architecture and dataset size. Across all other metrics - precision, specificity, F1-score, and ROC-AUC - the F1-optimised variant provided better results. Compute time was similar for both: 60.3 minutes for the F1-optimised model and 60.1 minutes for the recall-optimised one.

TABLE II. METRIC RESULTS OF MODELS WITH BEST HYPERPARAMETERS AFTER GRID SEARCH ON TEST SET

Model	**Recall**	**Precision**	**Specificity**	**F1-score**	**ROC-AUC**	**Compute time (minutes)**
Baseline CNN – F1 optimized	87.7 %	99.4 %	99.9 %	93.2 %	99.5 %	60.3
Baseline CNN – Recall optimized	87.7 %	96.1 %	99.0 %	91.7 %	99.2 %	60.1
ResNet18 – F1 optimized	89.8 %	100.0 %	100.0 %	94.6 %	98.1 %	117.9
ResNet18 – Recall optimized	91.0 %	95.2 %	98.7 %	93.1 %	99.3 %	117.6

3) Result comparison for ResNet18 model for F1-score and Recall search optimisation

A similar pattern is observed with ResNet18. The F1-optimised variant achieved better overall performance metrics. The recall-optimised variant produced slightly fewer missed seizures (35 vs 40 false negatives), but at the cost of considerably more false positives (18 vs 0). Compute time for ResNet18 variants was similar to each other, but approximately double that of the Baseline CNN models. The difference of five missed seizures between the two ResNet18 variants carries potential clinical significance - in a medical setting, each undetected seizure represents a risk to the patient [2] However, this modest gain in recall comes at the cost of 18 additional false positives, which may lead to unnecessary clinical interventions. The choice between the two variants therefore depends on the acceptable trade-off between false negatives and false positives in a given clinical context. The ResNet18-F1 model achieving perfect precision (100%) and zero false positives likely reflects the model being conservative - predicting seizure only when the probability is well above the 0.5 threshold. This is a consistent outcome of F1 optimisation combined with a relatively high positive weight value: the model learns to be selective about positive predictions rather than flagging uncertain cases as seizures. The result is zero false alarms, at the cost of 40 missed seizures - five more than the recall-optimised model. The results for all model variants were presented in the Confusion Matrices shown in Fig. 3.

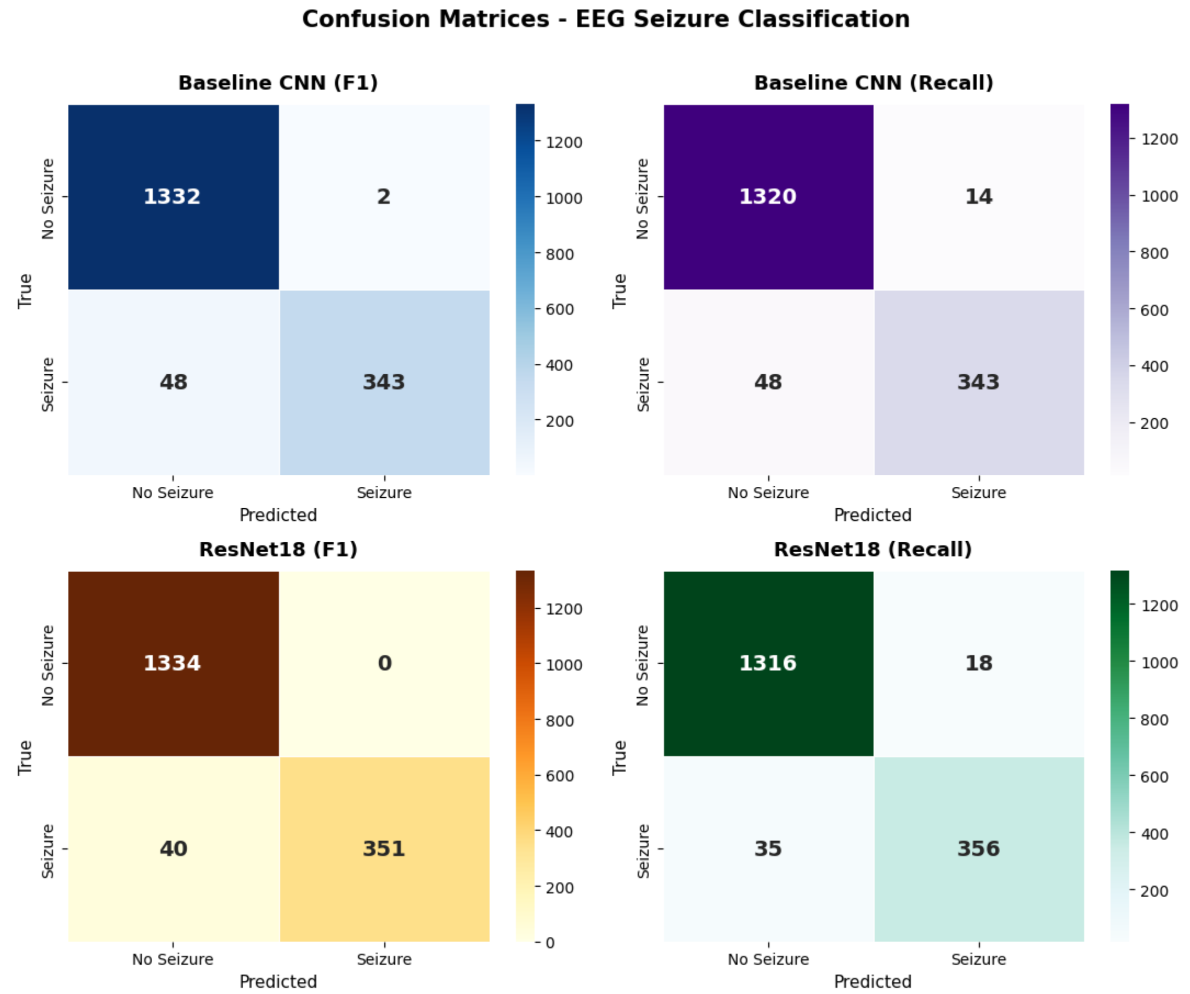

Fig. 3. Confusion Matrices for the test datasets across different models: Baseline CNN, F1 and Recall optimised, ResNet18, F1 and Recall optimised

4) Model generalisation

The generalisation ability of each model was assessed by comparing cross-validation scores obtained during hyperparameter search against final test set performance. For the F1-optimised models, the gap between CV and test scores was small - approximately 0.03 for both the Baseline CNN and ResNet18 - indicating stable generalisation to unseen data. For the recall-optimised variants, the gap was more pronounced, particularly for ResNet18, where the CV recall was noticeably higher than the test recall. This suggests that optimising for recall during the search may have led to hyperparameters that were better tuned to the training data distribution than to the unseen test set. Given the relatively modest dataset size, this is somewhat expected, as recall depends entirely on correctly identifying positive cases and is therefore more sensitive to distributional differences between training folds and the test set.

V. Conclusion

This work compared two deep learning architectures for automated epileptic seizure detection from EEG signals: a three-block Baseline CNN and ResNet18, each trained under two optimization strategies targeting F1-score and recall separately, using the UCI Epileptic Seizure Recognition Dataset.

ResNet18 optimised for F1-score achieved the strongest overall results, with an F1-score of 94.6%, perfect precision, and zero false positives. The Baseline CNN remained competitive with an F1-score of 93.2%, at roughly half the computational cost. The performance gap between the two architectures was modest, which suggests that added complexity does not always translate to proportionally better results.

Recall optimisation had mixed outcomes across both models. For the Baseline CNN it made no meaningful difference to sensitivity while increasing false positives sevenfold. For ResNet18 it reduced missed seizures by five compared to F1-optimised variant (35 to 40), but introduced 18 false positives. Whether this trade-off is acceptable depends entirely on the clinical context - in practice, the cost of a missed seizure and the cost of a false alarm are not equal.

The primary limitation of this work is the nature of UCI data itself. Each sample is one isolated 1-second window from single EEG channel, so the models have no access to temporal context or spatial information across brain regions. Future work could evaluate these architectures on a more clinically realistic dataset such as CHB-MIT, which provides longer continuous multi-channel recording. A second direction is to replace the general-purpose CNN models used in this work with EEG-specific deep learning models available through libraries such as Braindecode [15], specifically designed for raw EEG signal decoding, such as EEGNet and ShallowConvNet.

REFERENCES

[1] World Health Organization. Epilepsy Fact Sheet. Geneva: WHO; 2024. Available: https://www.who.int/news-room/fact-sheets/detail/epilepsy

[2] Benbadis, S.R.; Beniczky, S.; Bertram, E.; MacIver, S.; Moshé, S.L. The role of EEG in patients with suspected epilepsy. Epileptic Disorders 2020, 22, 143–155. doi: 10.1684/epd.2020.1151

[3] Türk, Ö.; Özerdem, M.S. Epilepsy Detection by Using Scalogram Based Convolutional Neural Network from EEG Signals. Brain Sciences 2019, 9, 115. doi: 10.3390/brainsci9050115

[4] Oto, M. The misdiagnosis of epilepsy: Appraising risks and managing uncertainty. Seizure 2017, 44, 143–146. doi: 10.1016/j.seizure.2016.11.029

[5] Shoeibi, A.; Khodatars, M.; Ghassemi, N.; Jafari, M.; Moridian, P.; Alizadehsani, R.; Panahiazar, M.; Khozeimeh, F.; Zare, A.; Hosseini-Nejad, H.; et al. Epileptic Seizures Detection Using Deep Learning Techniques: A Review. *Int. J. Environ. Res. Public Health* 2021, 18, 5780. doi: 10.3390/ijerph18115780

[6] Liu, G.; Xiao, R.; Xu, L.; Cai, J. Minireview of Epilepsy Detection Techniques Based on Electroencephalogram Signals. *Frontiers in Systems Neuroscience* 2021, 15, 685387. doi: 10.3389/fnsys.2021.685387

[7] Atlam, H.F.; Aderibigbe, G.E.; Nadeem, M.S. Effective Epileptic Seizure Detection with Hybrid Feature Selection and SMOTE-Based Data Balancing Using SVM Classifier. *Appl. Sci.* 2025, 15, 4690. doi: 10.3390/app15094690

[8] Bizopoulos, P.; Lambrou, G.I.; Koutsouris, D. Signal2Image Modules in Deep Neural Networks for EEG Classification. In *Proc. 41st Annu. Int. Conf. IEEE Engineering in Medicine and Biology Society (EMBC)*, Berlin, Germany, 2019, pp. 702–705. doi: 10.1109/EMBC.2019.8856620

[9] Dev, A.; Fouda, M.M.; Fadlullah, Z.M. Automated EEG Epileptic Seizure Detection Exploiting Continuous Morlet Wavelet Transform Scalograms and Hybrid Deep Learning Models. In *Proc. IEEE Int. Conf. Internet of Things and Intelligence Systems (IoTaIS)*, Bali, Indonesia, 2023, pp. 182–188. doi: 10.1109/IoTaIS60147.2023.10346065

[10] Ramírez-Zúñiga, C.; et al. Data leakage in deep learning studies of translational EEG. *PLOS Computational Biology* 2024. doi: 10.1371/journal.pcbi.1012001

[11] Rashid, H.U. *Epileptic Seizure Recognition* [Dataset]. Kaggle, 2018. Available: https://www.kaggle.com/datasets/harunshimanto/epileptic-seizure-recognition

[12] He, K.; Zhang, X.; Ren, S.; Sun, J. Deep Residual Learning for Image Recognition. In *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016, pp. 770–778. doi: 10.1109/CVPR.2016.90

[13] B. J. Kim, H. Choi, H. Jang, D. Lee, and S. W. Kim, "How to Use Dropout Correctly on Residual Networks with Batch Normalization," in *Proc. 39th Conf. Uncertainty in Artificial Intelligence (UAI)*, PMLR, vol. 216, pp. 1058–1067, 2023

[14] Akiba, T.; Sano, S.; Yanase, T.; Ohta, T.; Koyama, M. Optuna: A Next-generation Hyperparameter Optimization Framework. In *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery & Data Mining*, Anchorage, AK, USA, 2019, pp. 2623–2631. doi: 10.1145/3292500.3330701

[15] Schirrmeister, R.T.; Springenberg, J.T.; Fiederer, L.D.J.; Glasstetter, M.; Eggensperger, K.; Tangermann, M.; Hutter, F.; Burgard, W.; Ball, T. Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping* 2017, 38, 5391–5420. doi: 10.1002/hbm.23730